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The Editor
Nature

Dear Editor,

We submit our manuscript entitled “Metabolic buckling links growth scaling to scoliotic curve diversity” for consideration as an Article in Nature.

Adolescent idiopathic scoliosis (AIS) affects 3% of children worldwide, yet after more than a century of investigation, its cause remains unknown. The defining puzzle is temporal: 87% of scoliotic curves emerge within six months of peak height velocity — a window spanning less than 2% of skeletal development. No existing theory explains this precision, nor why the deformity is invariably three-dimensional, manifests in six distinct Lenke curve patterns, or disproportionately affects girls 10:1.

Our work resolves all four puzzles from a single physical principle. We demonstrate that the metabolic power required to maintain spinal alignment scales as L^4 (spinal length to the fourth power), while the metabolic supply during rapid growth scales as L^2 . This creates a transient energy deficit window during the adolescent growth spurt that triggers sequential failure of the most metabolically expensive mechanosensory proteins — a cascade we verify using structural analysis of 47 proteins via AlphaFold. The framework makes quantitative, falsifiable predictions for each Lenke curve type, the female prevalence, and the characteristic three-dimensionality of the deformity.

This work is significant for several reasons:

1. It provides the first unified pathogenetic theory for AIS, subsuming rather than competing with established hypotheses (melatonin, genetics, circadian desynchrony).
2. It introduces a novel metric (Conformational Ordering Index) that links protein structural biology to disease susceptibility through thermodynamic principles.
3. It identifies metabolic rescue as a rational therapeutic strategy — potentially transforming AIS management from reactive mechanical correction to proactive biochemical prevention.
4. The framework connects biomechanics, statistical physics, and developmental biology in a manner that will be of broad interest to Nature’s interdisciplinary readership.

This manuscript has not been submitted elsewhere and all authors have approved the sub-

mission. We declare no competing interests. Simulation code and AlphaFold analysis data are publicly available at <https://github.com/spinalmodes>.

We suggest the following potential reviewers with relevant expertise:

- Prof. Stuart Weinstein (University of Iowa) — clinical AIS epidemiology
- Prof. Alain Moreau (Université de Montréal) — AIS molecular biology
- Prof. M. Gail Faulkner (University of Alberta) — spinal biomechanics
- Prof. Geoffrey West (Santa Fe Institute) — allometric scaling in biology

We believe our findings will be of great interest to Nature’s broad readership, bridging fundamental physics and one of the most common unsolved problems in paediatric orthopaedics.

Yours sincerely,

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